FALCON 9

First-stage landing success analysis

Data collection, visual analytics and classification

Technical final report



Executive summary

90

Falcon 9 launches

66.7%

Course landing success

83.33%

Decision-tree test accuracy

What the data shows

Landing outcomes improve over the historical period, but vary by orbit, launch site and mission mix.

What the model supports

The reported tree correctly classifies 15 of 18 test launches. The small test set limits confidence in generalization.

EDA: IBM course dataset (90 launches). Model accuracy: reported lab answer.

Introduction and business question

Business context

Recovering a first stage creates the opportunity for reuse. Estimating landing outcomes can inform launch economics and risk discussions.

Analytical question

How do mission characteristics relate to first-stage landing success, and can a classifier predict the course label?

Target and boundary

Class = 1 follows the course landing rule. Launch mission success, physical recovery and economic reuse are distinct outcomes.

Decision use

Use descriptive findings and model scores as evidence for further analysis. This study does not estimate a launch price or a causal saving.

Scope: historical course data, primarily 2010–2020.

Data sources and evidence boundaries

Analysis	Sample	Evidence used
EDA and modeling features	90 launches	dataset_part_2.csv, 18 columns
SQL exploration	101 records	Spacex.csv, original text labels
Folium and Dash	56 launches each	Separate course geo and dashboard CSVs
Classification outcomes	18 test records	Prior-chat answers, original run not available

Different samples and label conventions explain differences in counts and rates.
Map and dashboard screenshots are reconstructions, not the original notebook outputs.

All calculated results in this deck use the public IBM course CSV snapshots.

Data collection via the SpaceX API

01 Request and flatten	Request historical launch JSON and use pandas.json_normalize to create a tabular dataset.
02 Resolve references	Use rocket, payload, launchpad and core identifiers to retrieve booster, payload mass, orbit, coordinates and reuse fields.
03 Restrict the cohort	Remove Falcon 1 launches and retain the Falcon 9 course sample of 90 records.
04 Persist the data	Save a CSV snapshot with flight number, date, launch site, outcome and engineered fields. Record the extraction date and schema.

Methodology reported by the course workflow. Historical snapshot retained for reproducibility.

Web scraping of launch tables

Extraction

Request the historical page and parse its HTML with BeautifulSoup. Inspect the title, identify launch tables and iterate through their rows.

Fields retained

Flight number, launch date and time, booster version, launch site, payload, mass, orbit, customer and outcomes.

Cleaning

Remove citation markers and footnotes, normalize units, parse numeric payload mass and standardize dates. Preserve original outcome wording.

Reproducibility

Wikipedia tables change. Save the page snapshot and extraction code. Keep scraped SQL records separate from the 90-row API cohort.

Source page: “List of Falcon 9 and Falcon Heavy launches”, Wikipedia.

Data wrangling and target definition

Rule	Implementation / result
Missing payload mass	Impute with the observed mean for the course exercise
Missing landing pad	26 / 90 values missing. Preserve or explicitly encode missingness
Class = 1	True ASDS (41), True RTLS (14), True Ocean (5) = 60
Class = 0	Failure, unknown or no successful landing label = 30
Scope of the target	Success rate = 60 / 90 = 66.7%, rounded to 67%

A more specific recovery label would count ASDS + RTLS: 55 / 90 = 61.1%.
Successful ocean landing does not demonstrate hardware recovery for reuse.

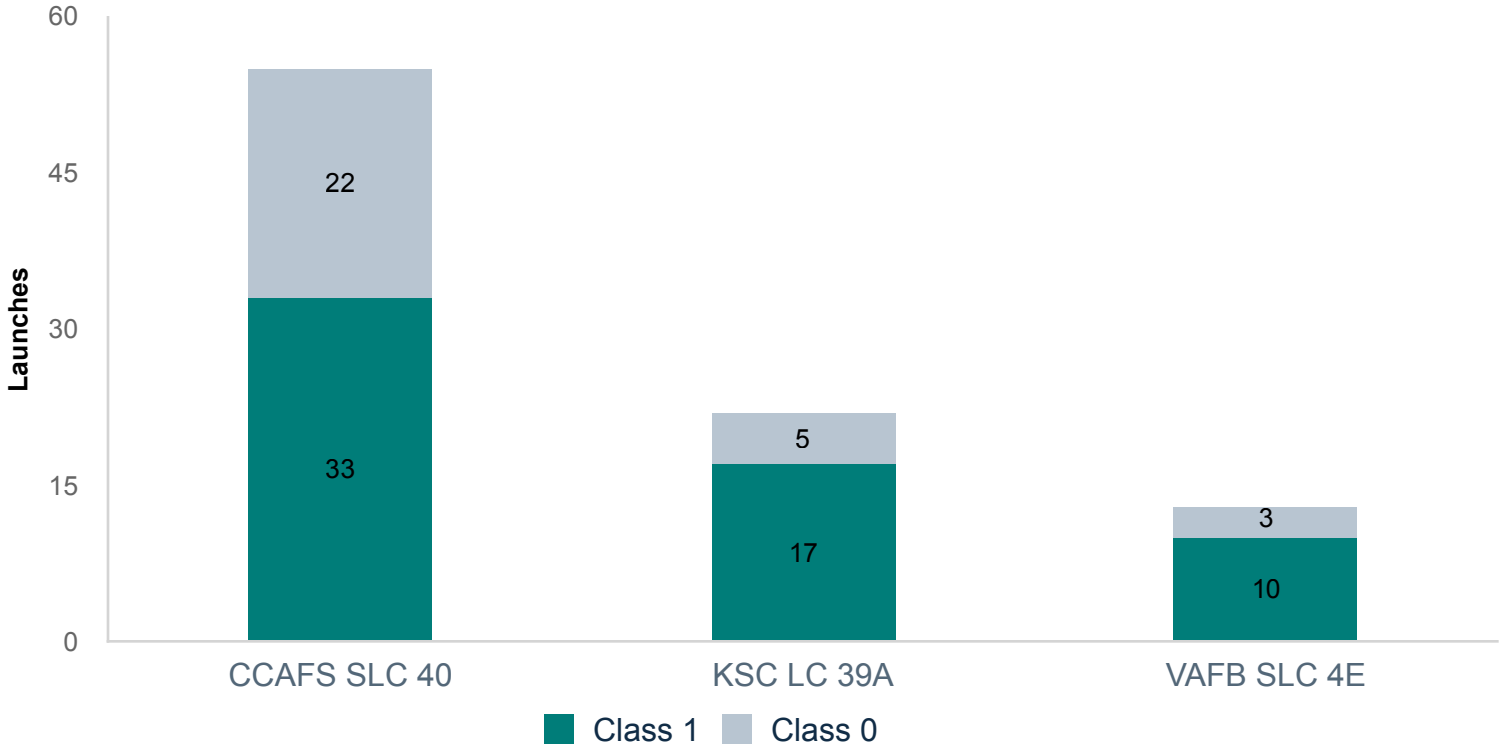
EDA sample: n = 90. Course Class includes successful ocean outcomes.

EDA and interactive visual analytics methodology

Visualization	Compare Class by launch site, orbit, payload mass, flight number and year. Show denominators with success rates.
SQL	Inspect DISTINCT values, aggregate payload mass, group outcomes and filter successful landings by date and booster.
Folium	Place launch coordinates on a map, cluster overlapping outcome markers and inspect coastal geography at different zoom levels.
Plotly Dash	Use a site dropdown and payload range slider to update the outcome pie chart and payload scatterplot through callbacks.

Methods cover descriptive patterns, geographic context and user-controlled filtering.

Launch sites differ in volume and success rate



55 launches

CCAFS SLC 40
33 / 55 = 60.0%

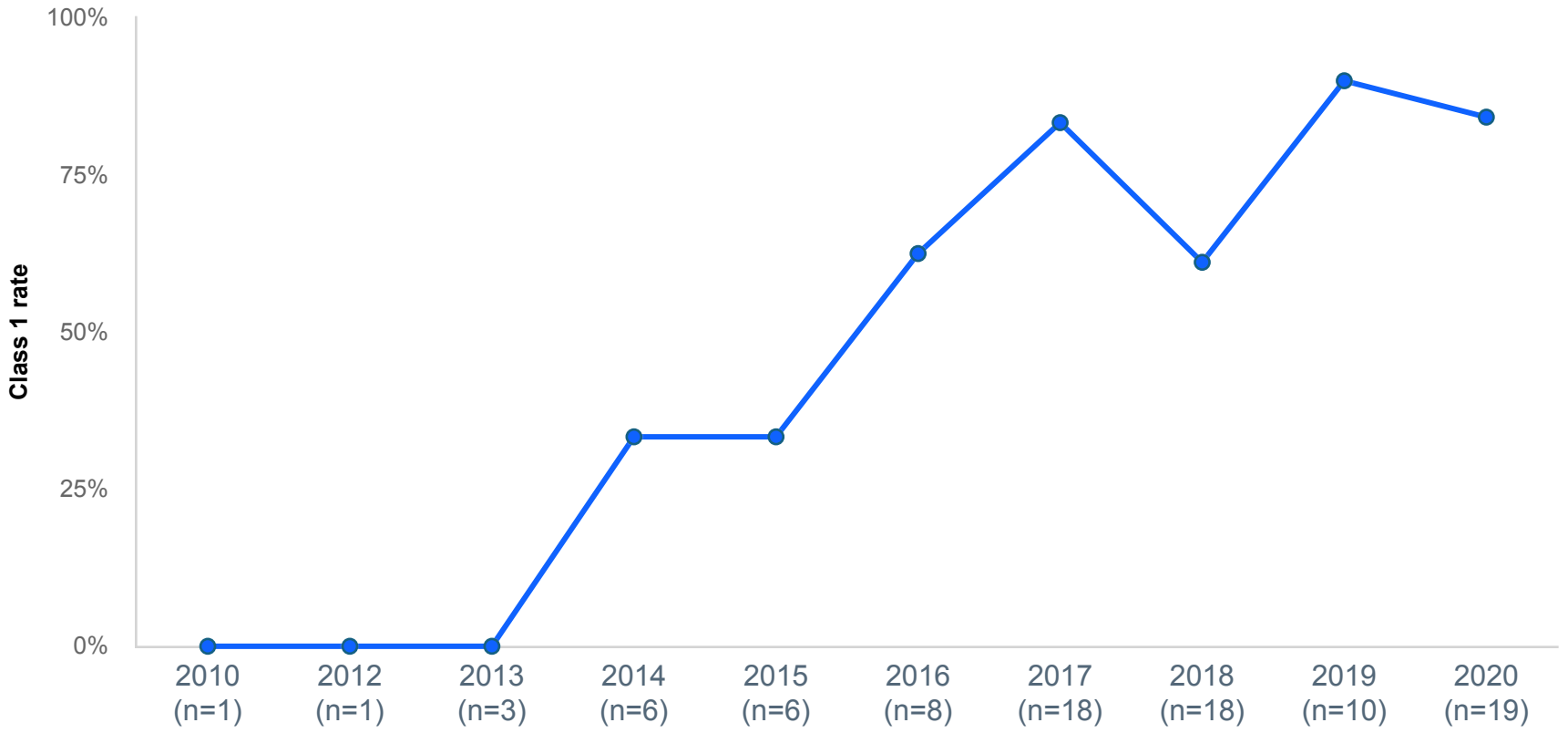
KSC LC 39A
17 / 22 = 77.3%

VAFB SLC 4E
10 / 13 = 76.9%

Site rates reflect different mission mixes and time periods. They do not isolate a causal site effect.

Recalculated from dataset_part_2.csv. n = 90.

Annual success rises, with a decline in 2018



83.3%

2017: 15 / 18

61.1%

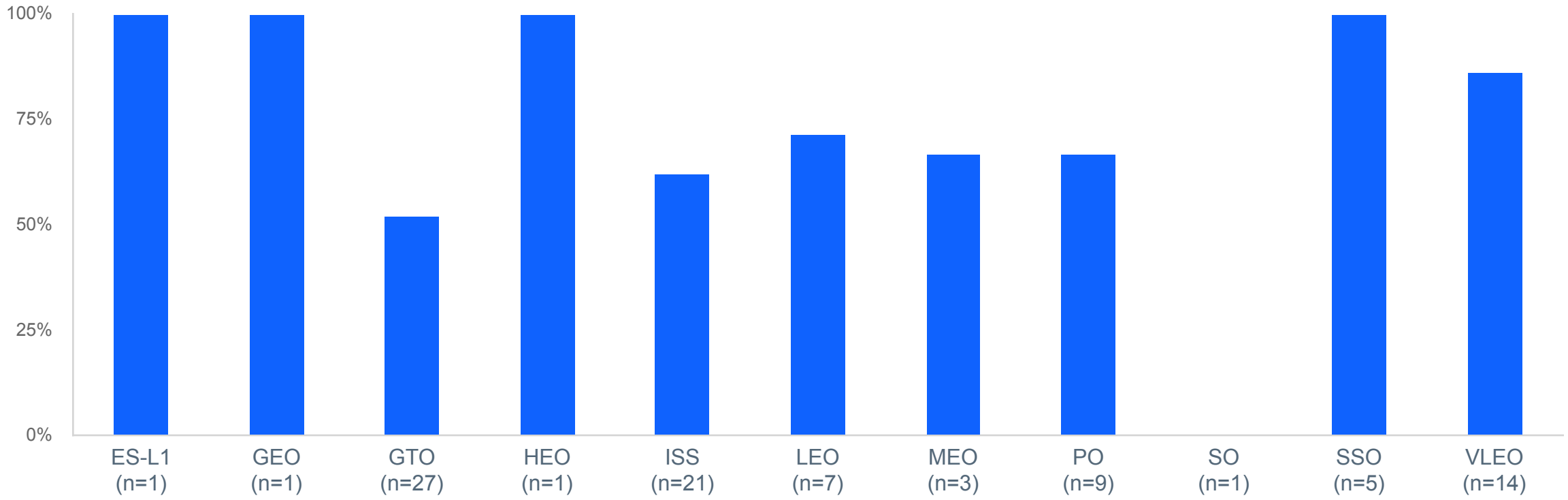
2018: 11 / 18

2020: 16 / 19
84.2%

Early cohorts are small. Hardware evolution and mission selection can confound the time trend.

EDA sample. Years without observations are omitted.

Orbit success rates need sample-size context

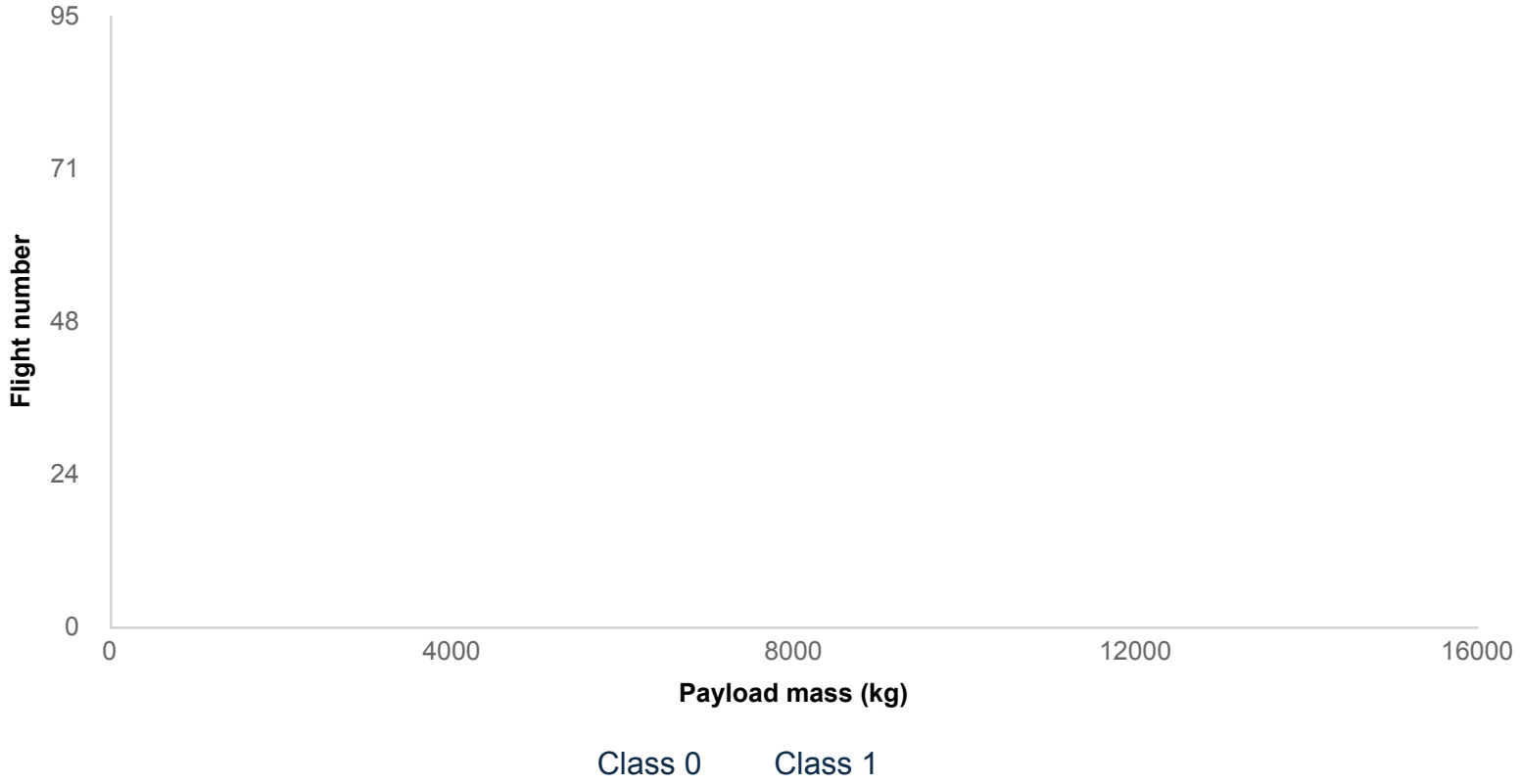


GTO: 14 / 27 = 51.9% **ISS: 13 / 21 = 61.9%** **VLEO: 12 / 14 = 85.7%**

GEO, HEO and ES-L1 each have one success in one launch. A 100% rate is weak evidence at n = 1.

EDA sample. Class 1 rate by orbit category.

Payload mass alone does not separate outcomes



Overlapping outcomes

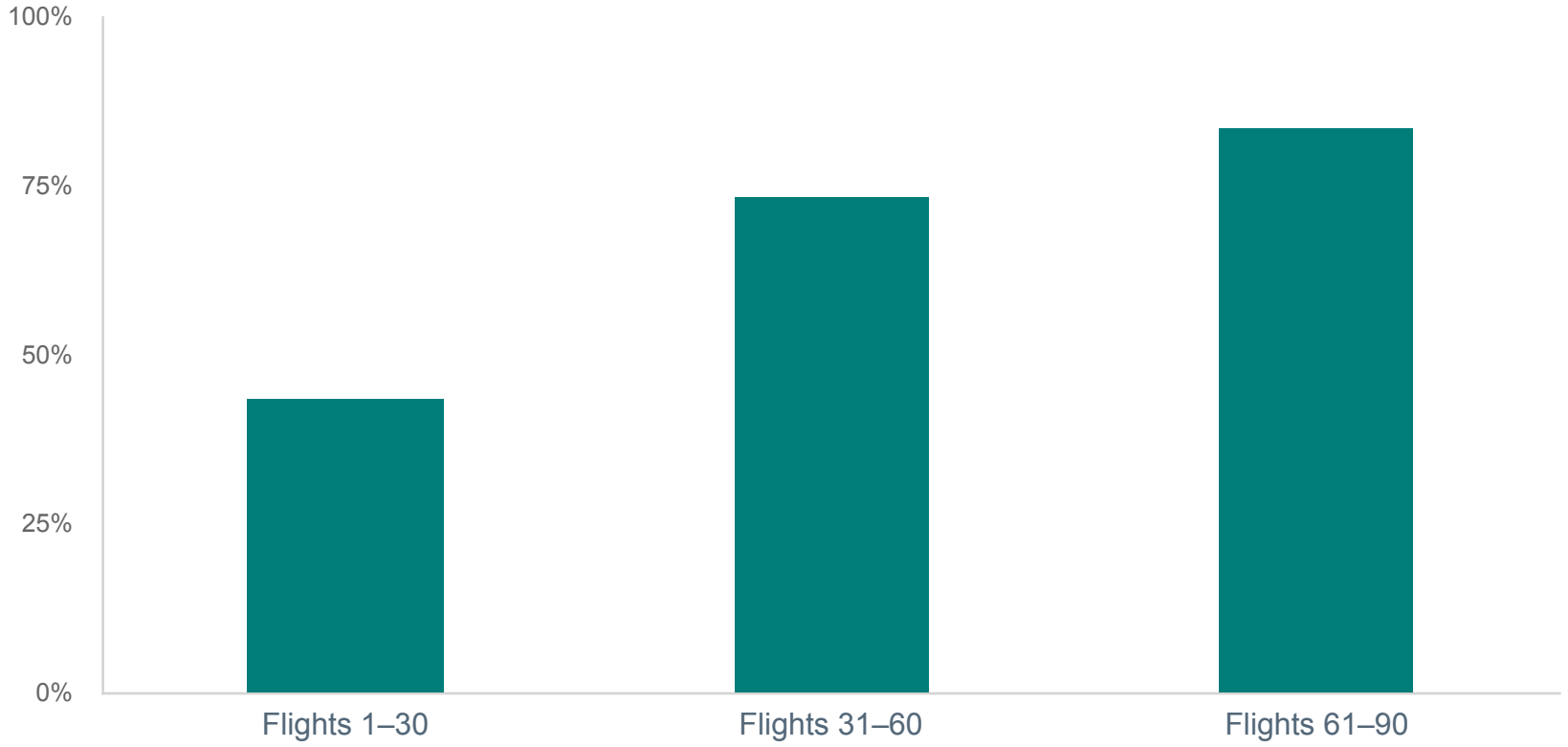
Successes and failures share payload ranges.

High payload flights mostly occur later in the sample.

Repeated values near 6,105 kg include mean-imputed observations.

EDA sample. PayloadMass includes course mean imputation.

Later flights contain more successful outcomes



Flights 1–30
13 / 30 = 43.3%

Flights 31–60
22 / 30 = 73.3%

Flights 61–90
25 / 30 = 83.3%

Flight number can encode operational maturity. Test on later launches to assess future performance.

EDA sample. FlightNumber is chronological within the retained cohort.

SQL methodology and reproducible queries

Site coverage

```
SELECT Launch_Site, COUNT(*) AS n
FROM SPACEXTBL
GROUP BY Launch_Site
```

NASA CRS payload

```
SELECT SUM(PAYLOAD_MASS__KG_)
FROM SPACEXTBL
WHERE Customer = 'NASA (CRS)'
```

First ground-pad success

```
SELECT MIN(Date)
FROM SPACEXTBL
WHERE TRIM(Landing_Outcome)
= 'Success (ground pad)'
```

Label checks

```
SELECT DISTINCT Launch_Site
FROM SPACEXTBL
```

Use TRIM before counting outcome labels.

SQL source: Spacex.csv, n = 101. Table name used below: SPACEXTBL.

SQL results quantify payloads and milestones

Question	Result	Query logic
Total NASA (CRS) payload	45,596 kg	SUM mass WHERE Customer = 'NASA (CRS)'
Average F9 v1.1 payload	2,534.67 kg	AVG mass WHERE Booster_Version LIKE 'F9 v1.1%'
First ground-pad success	22 Dec 2015	MIN date for 'Success (ground pad)'
Maximum payload mass	15,600 kg	MAX mass, then filter boosters at that mass
Minimum payload mass	0 kg	MIN mass. Zero needs source interpretation

Maximum-mass rows include B1048.4, B1049.4 and B1051.3. There are 12 matching records.

Computed from the 101-record SQL course snapshot.

SQL labels distinguish mission and landing outcomes

Launch-site label	Records
CCAFS LC-40	26
CCAFS SLC-40	34
KSC LC-39A	25
VAFB SLC-4E	16

99

Mission “Success” records after TRIM
1 failure in flight and 1 payload-status-unclear success

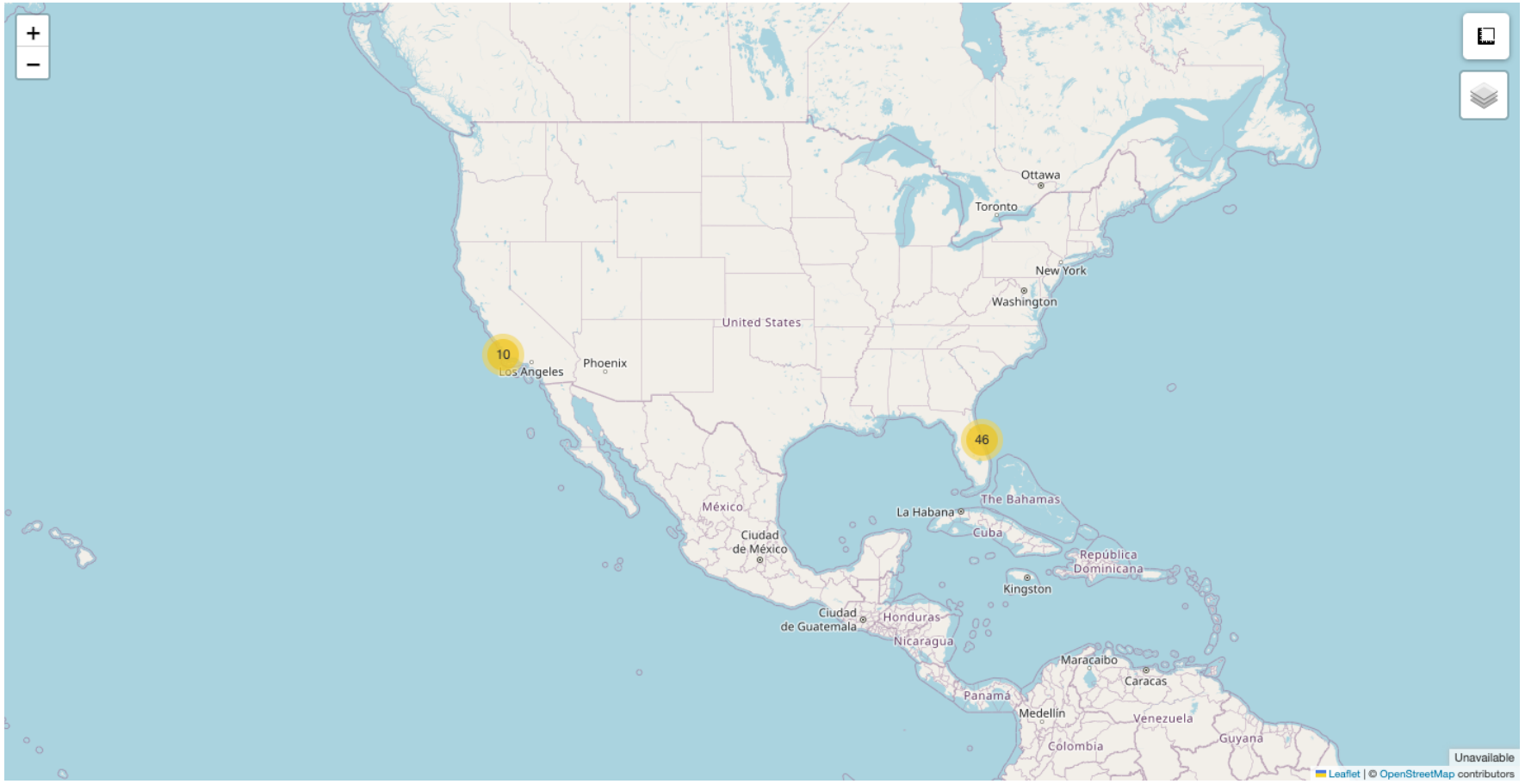
Computed from Spacex.csv. Leading/trailing whitespace normalized for outcome counts.

Folium interactive map methodology

Map layers	Plot latitude and longitude in Folium. Add site circles and launch markers with object.add_to(map) or map.add_child(object).
Outcome markers	Green marks Class 1 and red marks Class 0. Tooltips identify the source site and flight number.
Overlapping launches	MarkerCluster groups nearby markers. Zooming or expanding clusters reveals individual launch records.
Spatial inspection	MousePosition exposes coordinates. A measurement tool supports exploratory distances after checking coordinate accuracy.

Reconstructed from spacex_launch_geo.csv, n = 56.

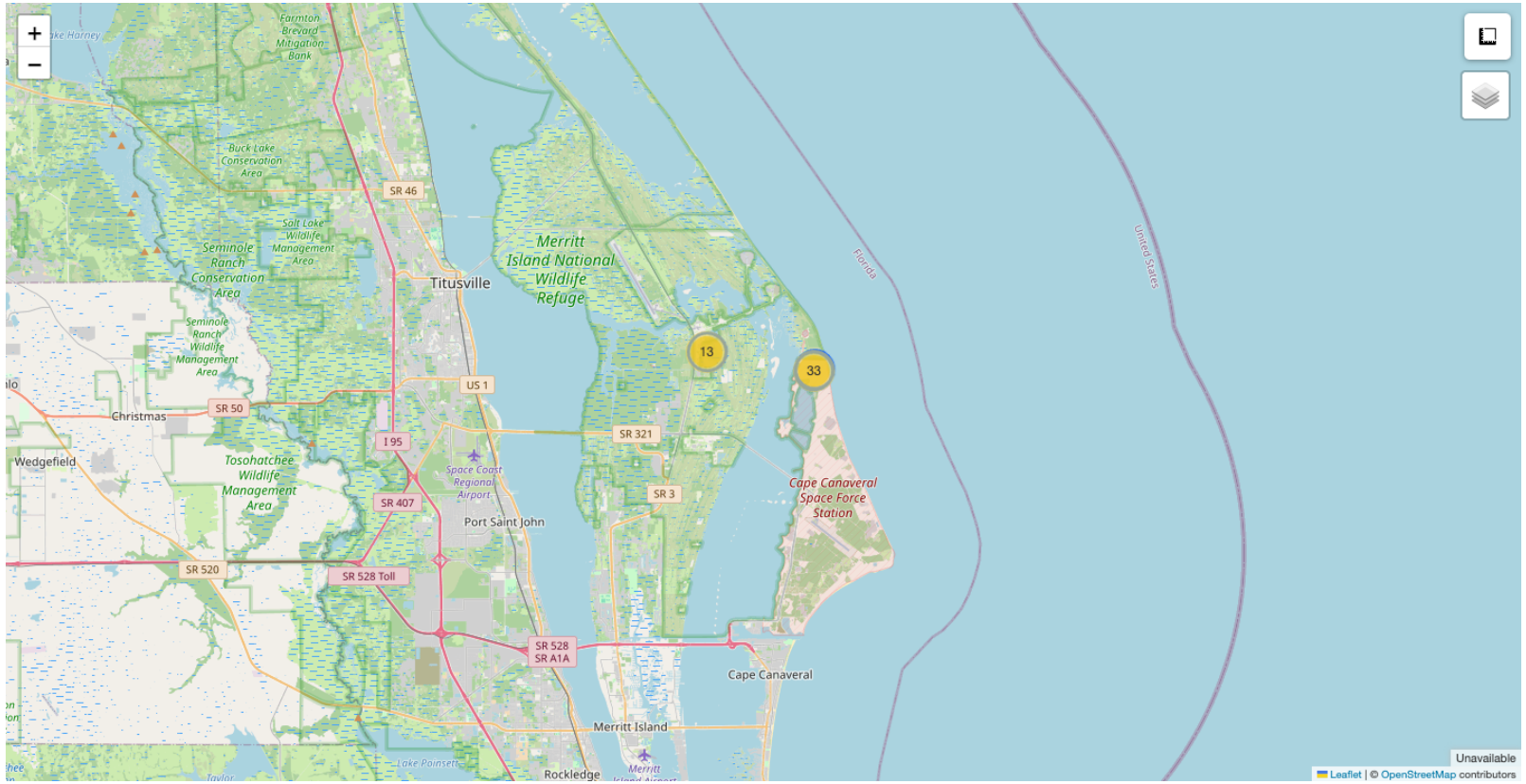
Folium map: launches cluster on two coasts



56 launch markers form two coastal clusters: 46 in Florida and 10 in California.

Reconstructed Folium output. Map tiles: © OpenStreetMap contributors.

Folium map: Florida sites require careful comparison



Geographic findings

Cape Canaveral sites sit near the coast.

LC-40 and SLC-40 labels refer to the same historical complex.

The course coordinates are approximate. Verify pad locations before reporting precise road or shoreline distances.

Coastal access is descriptive context. This map does not establish a causal link to landing success.

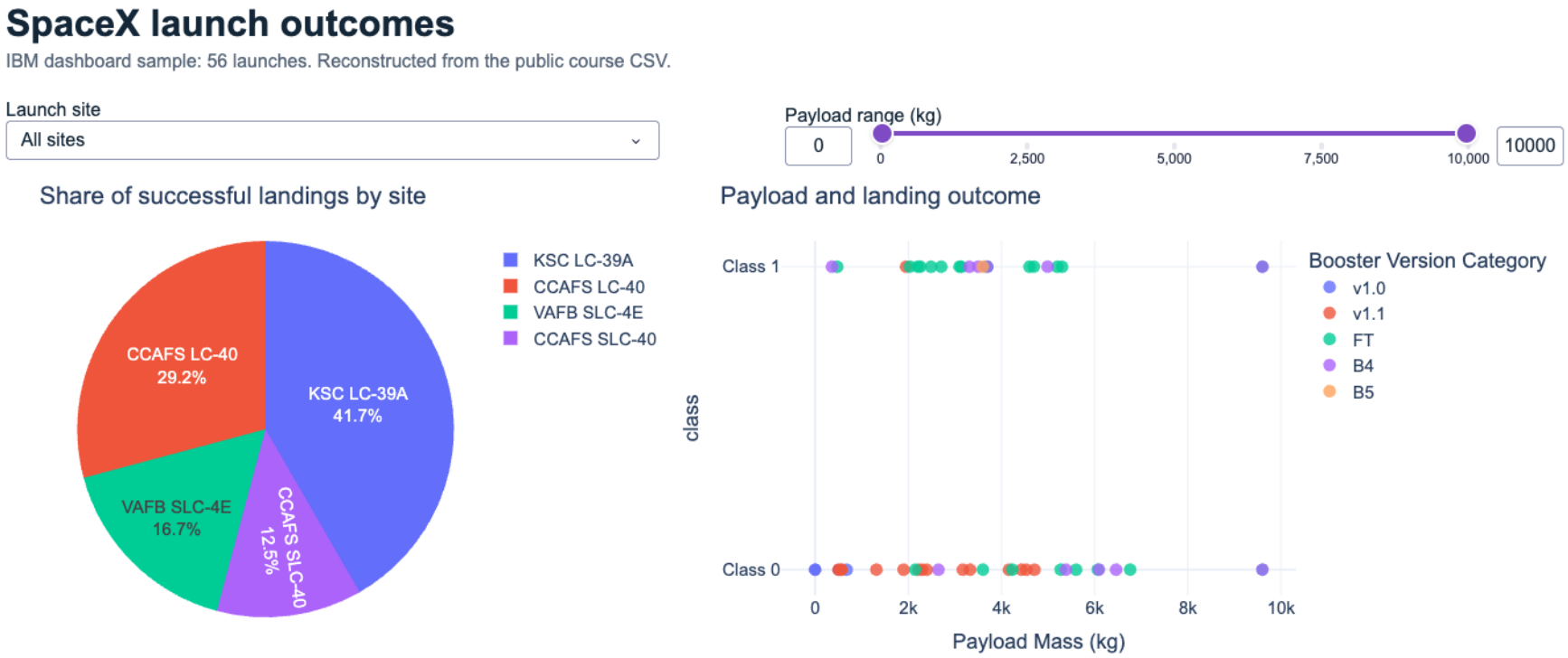
Reconstructed Folium view. Historical names and source coordinates retained.

Plotly Dash interaction design

Inputs	dcc.Dropdown options supplies site choices. A RangeSlider controls payload mass in kilograms.
Callbacks	Input component IDs and property names connect controls to a callback. Output IDs bind returned figures to dcc.Graph elements.
All-site pie chart	Sum Class by launch site to show each site’s share of successful outcomes. This is a success count share, not a site success rate.
Filtered scatterplot	Apply site and payload filters together. Plot payload against Class and color markers by booster-version category.

Reconstructed application uses the 56-row dashboard snapshot.

Dash dashboard: site and payload exploration



KSC LC-39A contributes 10 of 24 successful outcomes, or 41.7% of the all-site success pie.

Reconstructed Plotly Dash application. All sites, payload range 0–10,000 kg.

Dash results: success share differs from success rate

Source site label	Successes / launches	Site success rate	Share of 24 successes
CCAFS LC-40	7 / 26	26.9%	29.2%
CCAFS SLC-40	3 / 7	42.9%	12.5%
KSC LC-39A	10 / 13	76.9%	41.7%
VAFB SLC-4E	4 / 10	40.0%	16.7%

KSC LC-39A leads both measures in this snapshot. Payload scatter points overlap across outcomes, so no single payload cutoff explains success.

Dashboard snapshot n = 56. Class 1 total = 24.

Predictive features and preprocessing

Numeric and Boolean fields

FlightNumber, PayloadMass, Flights, GridFins, Reused, Legs, Block and ReusedCount.

Categorical fields

One-hot encode Orbit, LaunchSite, LandingPad and Serial. Cast the resulting features to numeric values.

Model matrix

Use the prepared features as X and Class as y. Standardization supports distance- and margin-based models such as KNN and SVM.

Leakage control

Fit imputation, scaling and encoding within each training fold. Confirm every predictor is available before launch, especially reuse-history fields.

83 encoded columns reported in the prior-chat lab answer.

Predictive analysis methodology

Split	Use an 80/20 train-test split of 90 rows: 72 training and 18 test records. Keep the test set outside hyperparameter selection.
Compare algorithms	Evaluate logistic regression, support vector machines, decision tree and k-nearest neighbors under the same split.
Tune on training data	Use GridSearchCV with cross-validation. Search SVM kernels and regularization, tree depth/criteria and KNN neighbor settings.
Evaluate once	Report held-out accuracy and a confusion matrix. Add class-specific precision/recall and compare against a majority-class baseline.

Course workflow summary. Recommended leakage controls stated separately.

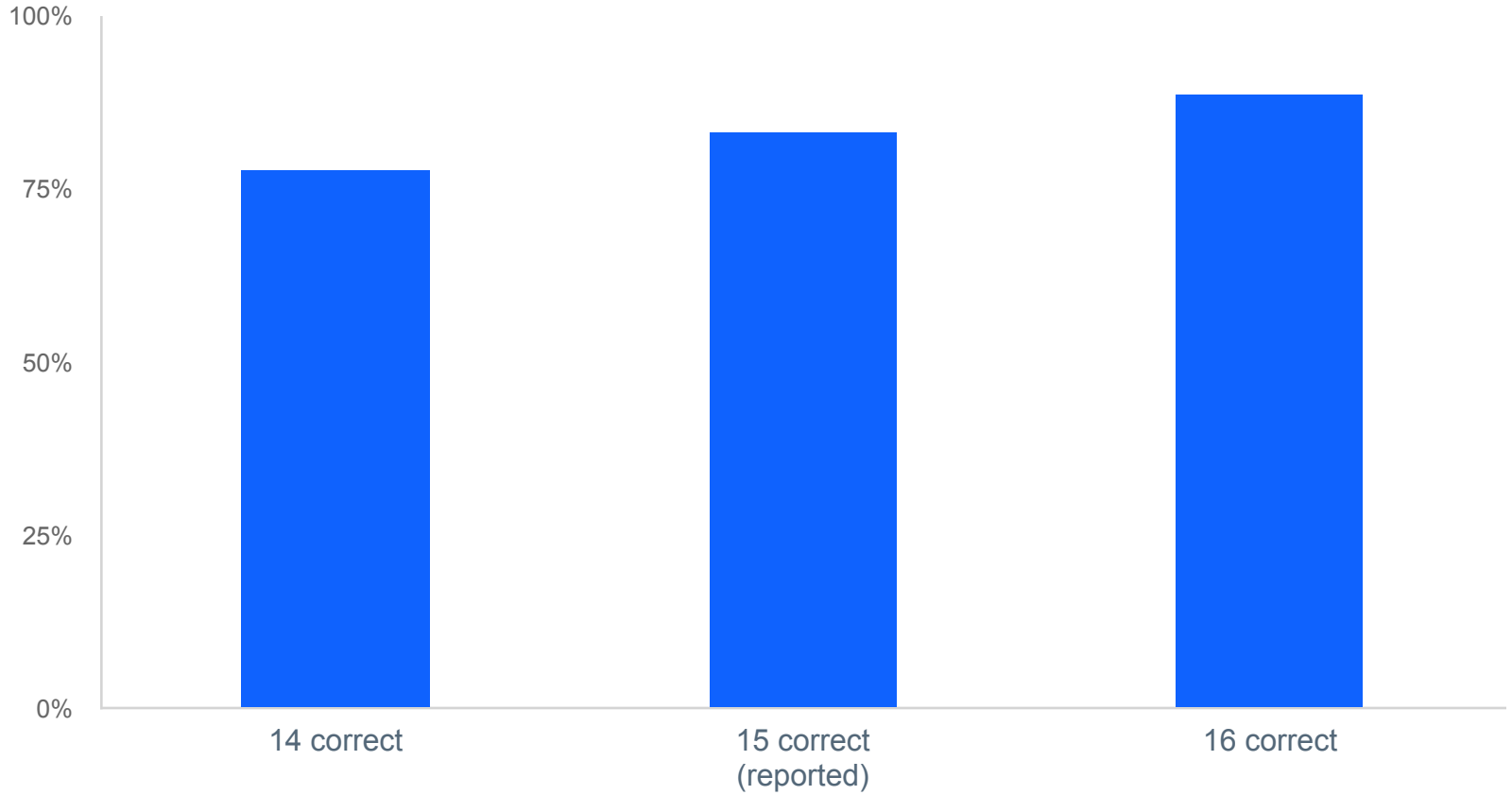
Classification results from the completed lab

Item / model	Available result	Interpretation
Held-out test sample	18 launches	Accuracy changes 5.56 points per launch
Support vector machine	Best kernel: rbf	Best validation kernel reported, score unavailable
Decision tree	Test accuracy: 83.33%	15 correct and 3 incorrect, implied by n = 18
Logistic regression	Score not supplied	No unsupported accuracy assigned
K-nearest neighbors	Score not supplied	No unsupported accuracy assigned

The available evidence does not establish a best overall model. An RBF kernel win is a within-SVM tuning result.

Model results reproduce the answers in the referenced conversation.

An 18-launch test set gives a coarse accuracy estimate



5.56 points

Accuracy change from one test prediction

60.8%–94.2%

Approximate 95% Wilson interval for 15 / 18

The interval assumes independent Bernoulli outcomes. Shared boosters and time effects can weaken that assumption.

Derived from the reported decision-tree accuracy and test-set size.

Innovative insights and next experiments

Recovery-specific target	Compare the course label (60/90) with ASDS-or-RTLS success (55/90). A business question about reuse needs a recovery-specific target.
Time-aware validation	Hold out the latest launches. Random splitting can mix early and mature hardware regimes and overstate future performance.
Booster-aware validation	Group folds by Serial to test generalization to unseen boosters. Compare results after removing high-cardinality Serial indicators.
Uncertainty in decisions	Calibrate predicted probabilities, then assess recovery value against recovery cost. No monetary result follows without cost and value inputs.

Interpretations and proposed extensions, not additional measured model results.

Conclusion

Descriptive findings

60 of 90 launches receive Class 1. Site and orbit rates vary, and later flight cohorts show higher success than early cohorts.

Interactive findings

Folium reveals coastal launch clusters. Dash demonstrates how site filters and payload ranges change the view of the 56-launch sample.

Predictive findings

The reported decision tree achieves 83.33% accuracy on 18 tests. RBF is the reported best SVM kernel. Full model ranking remains unavailable.

Practical next step

Publish the completed notebooks, validate a recovery-specific target and test generalization across launch dates and booster serials.

GitHub repository

https://github.com/falak711/Space-X

Repository contents

API and scraping notebooks
Wrangling and EDA notebooks
SQL queries and result outputs
Folium notebook and saved HTML

Model and report files

Dash application and requirements
Feature preparation and classifiers
Dataset provenance and README
Final presentation in PPTX and PDF

Sources and evidence notes

IBM course data

EDA / feature cohort	datasets/dataset_part_2.csv
SQL results	labs/module_2/data/Spacex.csv
Dashboard reconstruction	datasets/spacex_launch_dash.csv
Folium reconstruction	datasets/spacex_launch_geo.csv

Base URL: https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/

Model scores: referenced chat answers. API workflow: r-spacex/SpaceX-API.
Scraping source: Wikipedia launch list. Cover: NASA / Joel Kowsky.
Map: CARTO and OpenStreetMap contributors. Full source URLs appear in slide notes.

Grading coverage

Requirement	Slide(s)	Requirement	Slide(s)
GitHub repository link*	29	Folium map results	17–19
PDF presentation	This PDF	Dash results	20–22
Executive summary	2	Predictive results	25–26
Introduction	3	Conclusion	28
Collection and wrangling	5–7	Creativity	Charts + reconstructions
EDA / visual methods	8, 17, 20	Innovative insights	27
Predictive methods	23–24	EDA visualization results	9–13
SQL results	14–16	Sources and provenance	4, 30

15 requested criteria mapped to this report.